

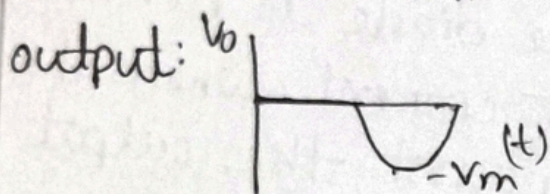
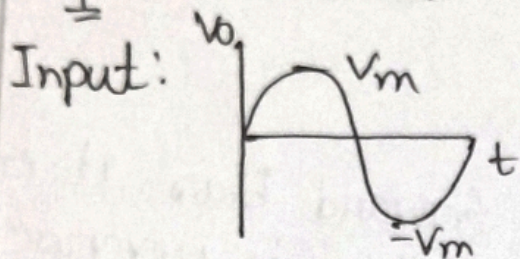
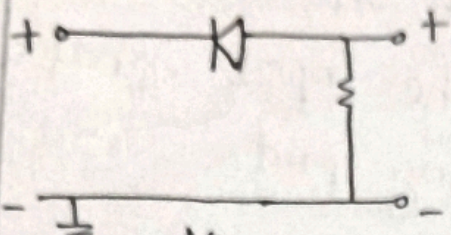
Definition of series Clipper: A series clipper is a diode circuit in which the diode is connected in series with the load resistor. It clips a part of the input signal without changing the remaining part.

Working principle:

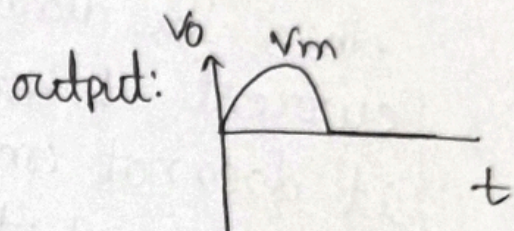
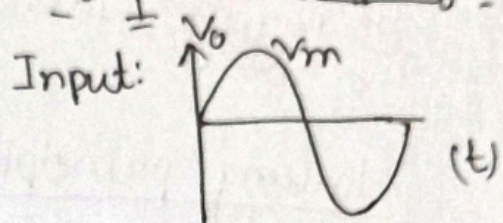
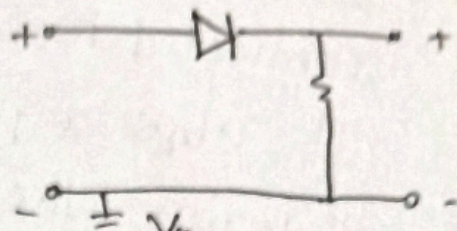
When the diode is forward biased it conducts current. When the diode is reverse biased it does not conduct current. When the diode does not conduct current the output voltage become zero. The direction of the diode decides which part of the signal will be clipped. If a DC battery added, the clipping level can be changed. This type is called a biased series clipper. When the diode is ON, output voltage is found using KVL. for a biased series clipper $V_o = V_i - V$ when the diode is ON. when the diode is off $V_o = 0$

Series Clipper:

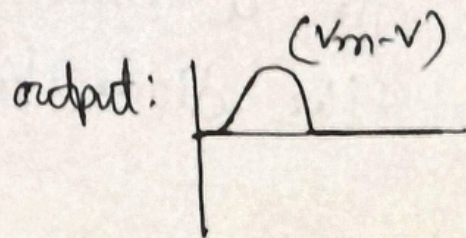
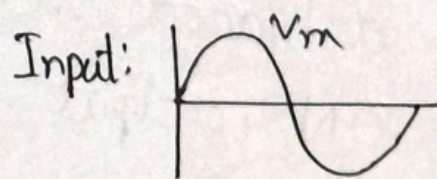
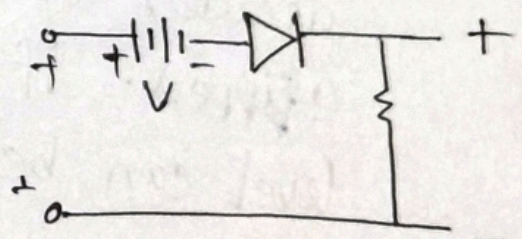
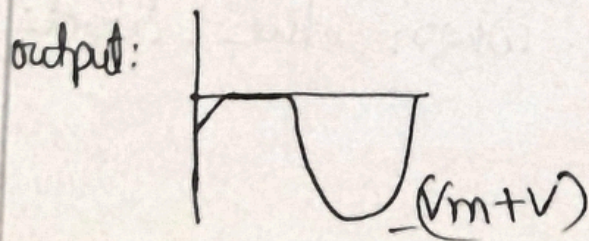
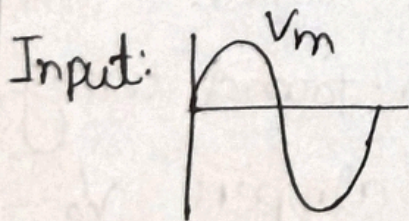
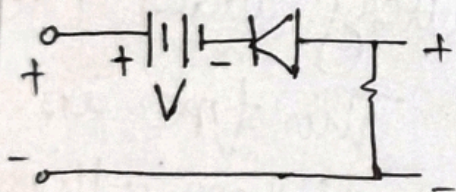
Positive:



Negative



Biased Series Clipper:-



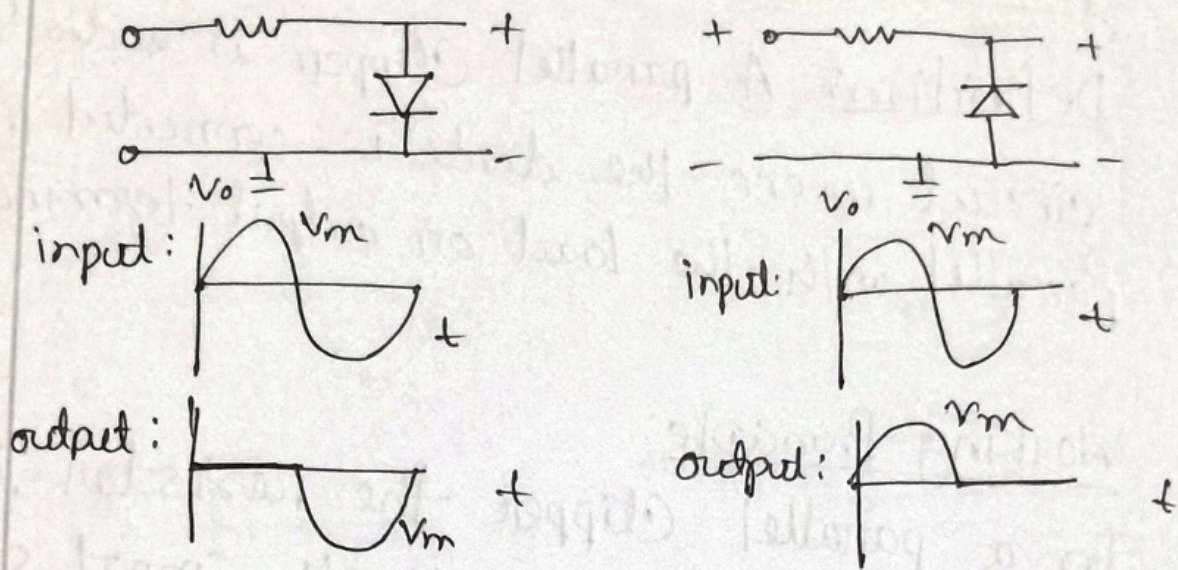
Parallel Clippers:

Definition: A parallel Clipper is a diode circuit where the diode is connected in parallel with the load or output terminal.

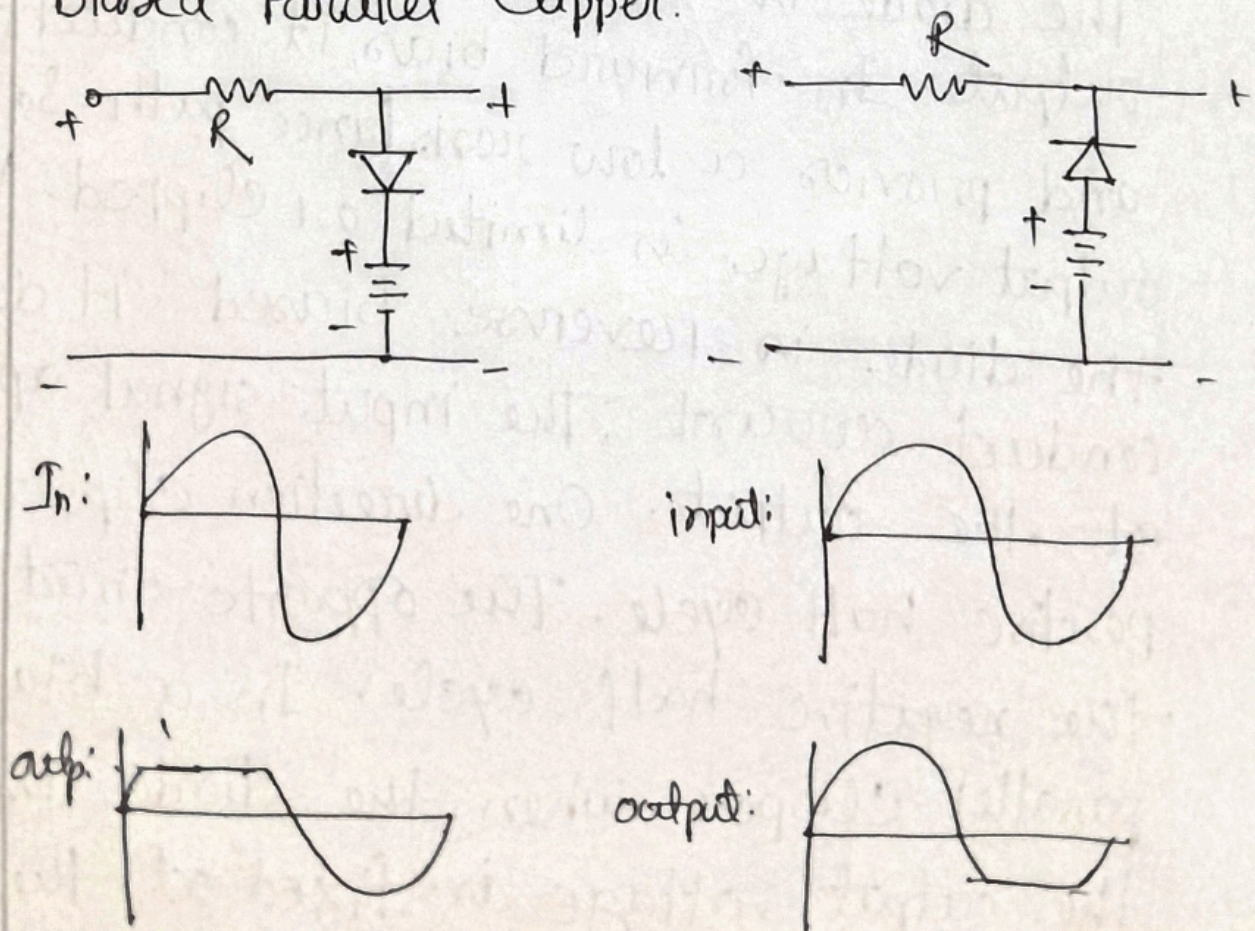
Working Principle:

In a parallel Clipper the resistor is connected in series with the input signal. The diode is connected parallel to the output. In forward bias, it conducts current and provides a low resistance path. So the output voltage is limited or clipped. When the diode is reverse biased, it does not conduct current, the input signal appears at the output. One direction clips the positive half cycle. The opposite direction clips the negative half cycle. In a biased parallel Clipper when the diode is on the output voltage is fixed at the clipping level. When the diode is OFF output voltage follows the input signal.

Simple Parallel Clipper



Biased Parallel Clipper:

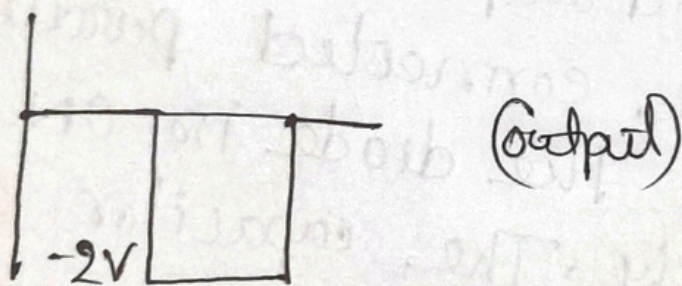
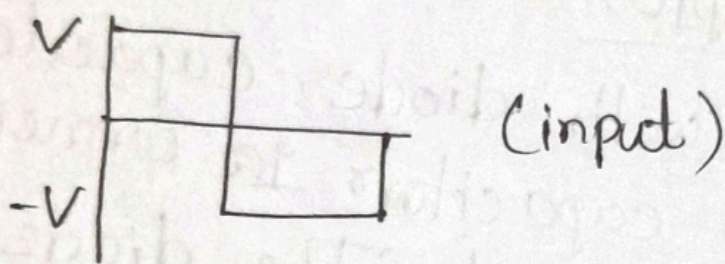
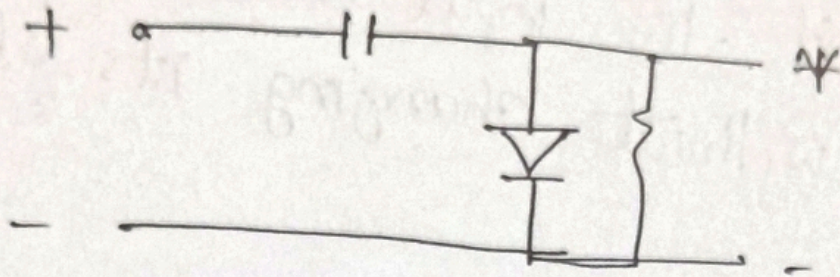


Clampper: A clamper is a diode circuit that shift the input waveform to a different DC level without changing its shape.

Working principle:

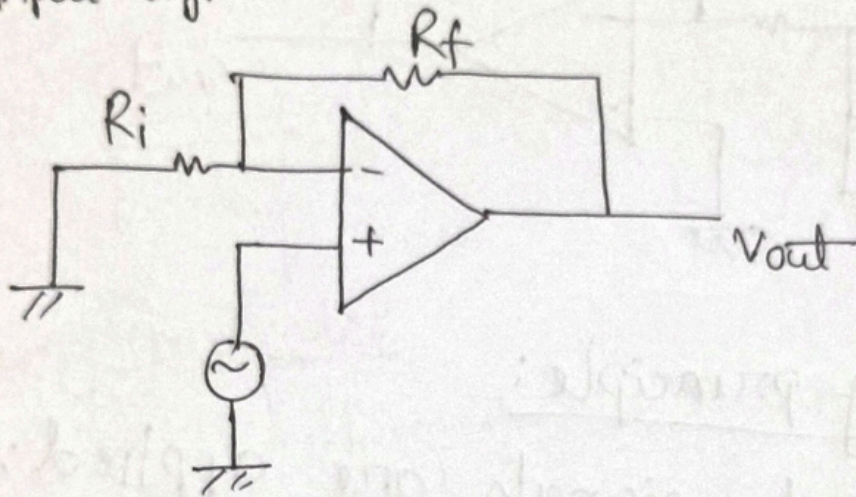
It is made with diode, capacitor and resistor. The capacitor is connected between the input and output. The diode and resistor are connected parallel to the output. When the diode is ON capacitor charges quickly. The capacitor charges up to a certain voltage level. When the diode is off the capacitor holds its charge. The stored capacitor voltage shifts the whole waveform upward or downward. The shape of the waveform remains the same. Only the DC level of the waveform changes. The output signal has the same total swing as the input signal.

Clipper: Clamper:



Non inverting op-amp: A non inverting op amp is an amplifier circuit where the input signal is applied to the non-inverting terminal of the op-amp.

The output signal has the same phase as the input signal.

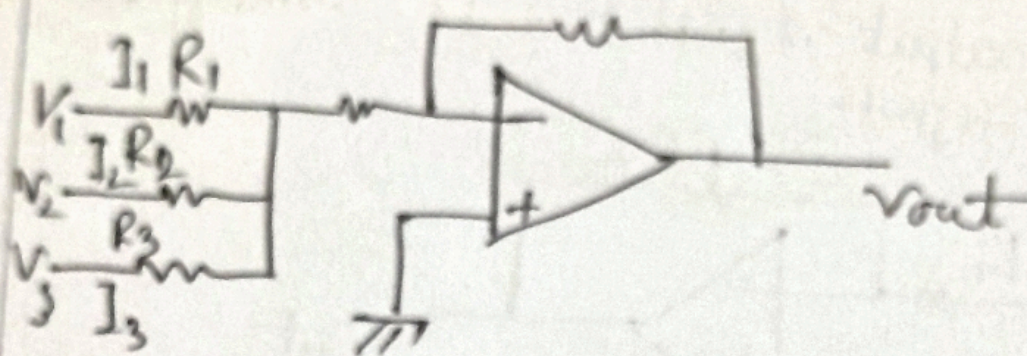


Working principle: The inverting terminal is connected to a feedback network. another resistor is connected between the output and the inverting terminal. Due to negative feedback the op-amp maintains both input terminals at nearly the same voltage. The output voltage follows and amplifies the input voltage. The gain depends on the feedback resistor and ground resistor.

$$\frac{V_{out}}{V_{in}} = \left(1 + \frac{R_f}{R_i}\right)$$

output is not inverted but amplified.

Inverting Adder: An inverting adder is an op-amp circuit that adds two or more input voltages. It is also called summing amplifier.



Working principle:

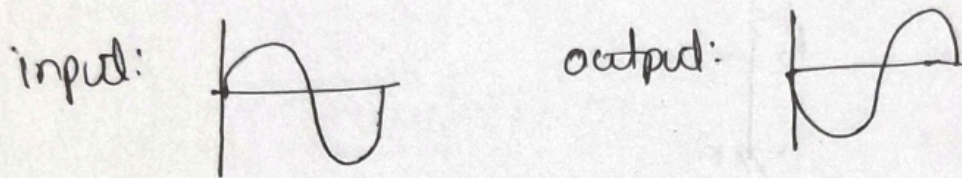
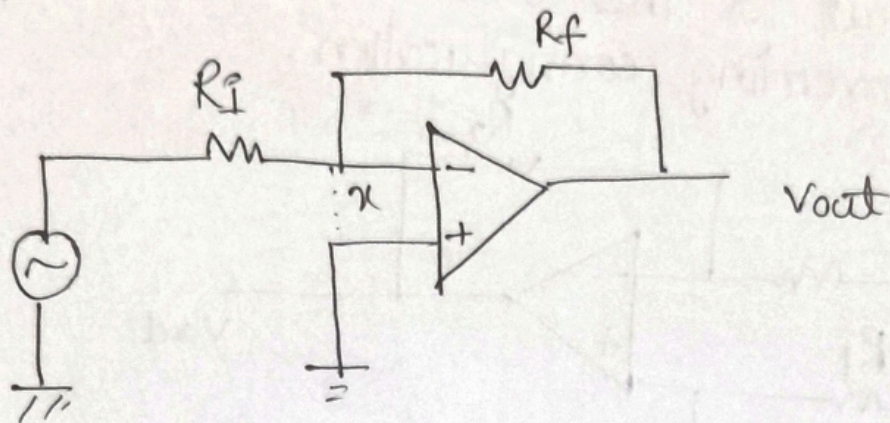
All input signals are applied to the inverting terminal through separate input resistors. The non-inverting terminal is connected to the ground.

All the input currents meet at the inverting terminal and flow through the feedback resistor. The op-amp converts the total current into output voltage.

The output is the negative sum of all input voltages. Output is inverted.

$$V_{out} = -(V_1 + V_2 + V_3)$$

Inverting OP-AMP → An inverting opamp is an amplifier circuit where the input signal is applied to the inverting terminal of the op-amp.



Working principle:

The non inverting terminal is connected to the ground. The output signal is opposite in phase to the input signal. That means the output is inverted by 180° .

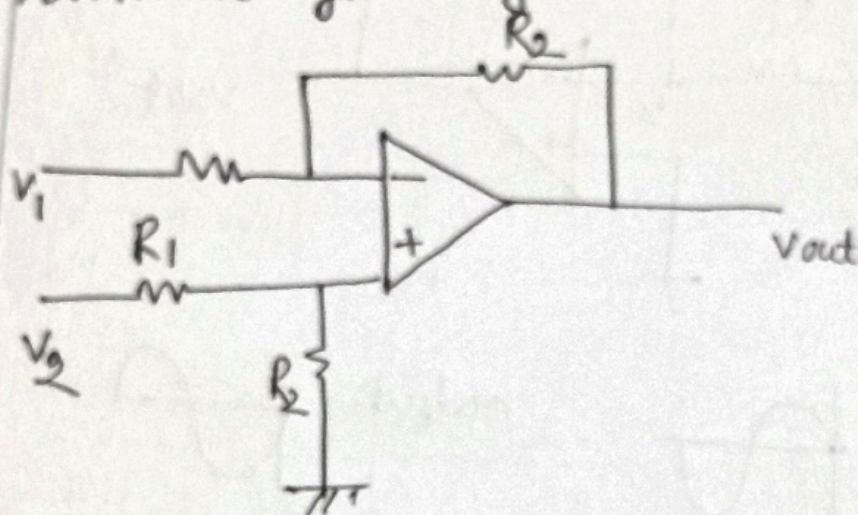
A feedback resistor is connected between the output, the inverting input. Current flows through the input resistor and feedback resistor.

The op-amp produces an output voltage,

The voltage gain
$$\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

negative sign shows the output is inverted.

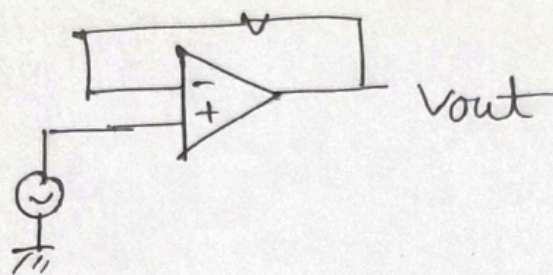
Subtractor: A circuit that takes the difference between two input signals. This opamp circuit is the combination of inverting and non inverting configuration.



Working principle:

(i) It subtracts one input voltage from another input voltage. It is also called difference amplifier. Two signals are applied. One signal is applied through the inverting path. Another signal is applied through the non-inverting amplifier path. The opamp compares the two input signals. The output depends on the difference between the two inputs. $V_o = E_1 - E_2$.

Voltage follower: A voltage follower is an op-amp circuit whose output voltage is equal to the input voltage. It is also called buffer amplifier.



Working principle: The input signal is applied to the non inverting terminal. The output is directly connected to the inverting terminal. Due to the negative feedback the op-amp makes the inverting voltage equal to the non-inverting terminal voltage. So the output voltage follows the input voltage. The voltage gain = $\frac{V_{out}}{E} = 1$

It has high input resistance, low output resistance.